LEFT-HAND vs RIGHT-HAND MOTOR IMAGERY WITH AN SVM - FULL EXPERIMENT REPORT

Folder: D:\trainsvm Written: 19 September 2026

This file explains, in plain English, everything that was done in this folder: what we tried, why, how, what came out, and what it means. You should be able to read it without any other document. Short definitions of technical words are at the end (section 9, GLOSSARY).

0. THE ONE-PARAGRAPH SUMMARY

The goal was to train an SVM (a type of machine-learning classifier) that reads brain signals (EEG) and says whether a person is IMAGINING moving the LEFT hand or the RIGHT hand. We worked on two datasets:

(1) A published dataset of 50 stroke patients (Liu et al., 2024). The paper claims up to 72 % accuracy. When we tested everything carefully and honestly, the best we could get was about 55 %. The main reason is that the authors' own code chooses its best result by looking at the test data, which inflates the score. Our honest numbers are the believable ones.

(2) Our own recordings from the Gelo BCI headband (4 electrodes). When tested the easy (wrong) way, the SVM looks excellent (78-98 %). When tested the correct way (on a recording the model has never seen), it scores 57-65 %, and a statistical test says even 65 % is not clearly better than guessing (p = 0.089). The reason is how the data was recorded: every file contains only left OR only right, so the model learns to recognise the recording session instead of the hand. The fix is to record left and right trials mixed in random order inside the same session.

"Chance" in this whole report is 50 %: with two classes, a coin flip gets 50 % right.

1. BACKGROUND - WHAT WE ARE TRYING TO DO

1.1 Motor imagery

When you imagine moving your LEFT hand, the rhythm of the brain area that controls it changes. That area is on the RIGHT side of the head (the brain is crossed over). An electrode called C4 sits above it. For the RIGHT hand, the area is on the LEFT side of the head, under electrode C3.

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The change we look for: the brain's "idling" rhythms get weaker over the active side.
  - mu rhythm   : 8-13 Hz
  - beta rhythm : 13-30 Hz
This weakening is called ERD (event-related desynchronisation).
So, in theory:
  - imagine LEFT hand  -> mu/beta power DROPS at C4 (right side of head)
  - imagine RIGHT hand -> mu/beta power DROPS at C3 (left side of head)
A classifier can learn this left/right difference. In practice the effect is small, it
differs a lot between people, and stroke patients may have it weakened or missing.
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1.2 SVM (Support Vector Machine)

```
An SVM takes a list of numbers per example (called "features", e.g. the power at C3 and
at C4) and learns a boundary that separates the two classes. The main settings are:
  - kernel : "linear" (a straight boundary) or "rbf" (a curved boundary)
  - C      : how strictly it tries to fit the training data (big C = fits tightly and
             risks memorising noise; small C = smoother boundary)
  - gamma  : only for rbf, how curvy the boundary may be
The SVM never sees raw EEG. It sees FEATURES that we compute from the EEG.
```

1.3 The single most important idea in this report: HONEST TESTING

To know how good a model is, you must test it on data it did NOT learn from. And you must not make ANY decision (which settings, which frequency band, which model) by looking at the test results. If you do, you are "peeking", and the score you report is too high.

We use cross-validation (CV): split the data into parts (folds), train on some, test on the rest, repeat so every part is tested once, then average.

Two more dangers that matter for our data:

- LEAKAGE: pieces of the same recording in both the training set and the test set. The model may simply recognise the piece it has already seen.

- CONFOUND: something other than left/right that also differs between the classes (for example, the day or the session they were recorded in). The model can learn that instead of the hand.

Whenever the report says a number "cannot be trusted", it is because of one of these.

PART A - THE PUBLISHED STROKE-PATIENT DATASET

2. THE PAPER AND THE DATA

2.1 The paper

Liu et al., "An EEG motor imagery dataset for brain computer interface in acute stroke patients", Scientific Data 11:131 (2024). The data is in D:\trainsvm\eeg_stroke.

2.2 What the data contains (what we found by reading the files)

- 50 acute stroke patients, one session each.
- 40 trials per patient: 20 left-hand, 20 right-hand, in ALTERNATING order.
- Each trial is 8 seconds: 2 s instruction -> 4 s motor imagery (patient watches a video and imagines the movement) -> 2 s break
- 500 samples per second (500 Hz), so one trial = 4000 samples.
- 33 channels:
  - 30 EEG electrodes; channel 18 (CPz) is the reference and is a flat constant, so 29 EEG channels are really usable
  - 2 eye channels (EOG)
  - 1 marker channel: 1 = instruction, 2 = imagery starts, 3 = break
- Two copies of the data:
  - raw .mat files (unfiltered, large DC offsets of about 10,000 microvolts)
  - .edf files (already filtered 0.5-40 Hz, stored as one continuous 320 s recording)
- The authors' MATLAB code is included (FBCSP_SVM.m, CSP_LDA.m, TSLDA_DGFMDM.m, TWFB_DGFMDM.m and helpers).

2.3 How the authors train their SVM (read from FBCSP_SVM.m, not just the paper)

Raw data IS used (the .mat files), with this processing:

1. Keep 29 EEG channels (drop reference, eye and marker channels).
2. Find the moment imagery starts (marker = 2). Cut 0 to 4 s after it.
3. Filter: remove 50 Hz mains noise, then keep 4-30 Hz (Butterworth order 2, one-way, "causal").
4. Features = FBCSP (Filter-Bank CSP):
   - split into 5 bands: 10-13, 13-16, 16-19, 19-22, 22-25 Hz
   - in each band, CSP finds 2 pairs of "spatial filters" (weighted mixes of channels) that best separate left from right
   - feature = log of the signal variance through each filter
   - 5 bands x 4 filters = 20 numbers per trial
5. Classifier: MATLAB fitcecoc with default settings = linear SVM, C = 1, no scaling.
6. Testing: 10 times, randomly pick 24 trials to train and 16 to test (60/40). Not balanced by class, no fixed random seed. Report the average.

Note: the PAPER says 8-30 Hz filtering and 10-fold cross-validation, but the CODE does 4-30 Hz and random 60/40 splits. The code is what actually ran.

```
Paper's reported accuracies (average of 50 patients):
  CSP + LDA ..................... 55.57 %
  FBCSP + SVM ................... 57.57 %
  TSLDA + DGFMDRM ............... 61.20 %
  TWFB + DGFMDRM ................ 72.21 %   (their best)
```

3. WHAT WE DID WITH THE STROKE DATA, STEP BY STEP

STEP A1 - Faithful copy of the authors' SVM in Python

```
Script : stroke_paper\train_fbcsp_svm.py
What   : exactly the pipeline in 2.3, translated from MATLAB to Python.
Problem: in 11 patients, 18 trials have the imagery marker at the very first sample,
         so the 4 s window runs past the end of the trial. Fixed the same way MATLAB
         does it: join all 40 trials into one continuous stream, then cut windows.
Result : 47.86 % average - basically chance, and well below the paper's 57.57 %.
```

STEP A2 - Try to improve the SVM

Script: stroke_paper\train_improved_svm.py

Changes (all standard good practice):

- use the "core" imagery period 0.5-3.5 s (skip the reaction delay at the start)
- scale the features (StandardScaler) before the SVM; SVMs need this
- choose C (0.001, 0.01, 0.1, 1, 10) by an INNER cross-validation that uses only training data ("nested CV"), so the choice never peeks at the test set
- FBCSP with 6 bands 8-30 Hz, keep the best 8 features (mutual information)
- new feature types: Riemannian "tangent space" (see glossary), on one band and on three bands (8-13, 13-20, 20-30 Hz)
- testing: stratified 10-fold CV (each fold has equal left/right)

Result: 51.9 % (FBCSP), 53.3 % (tangent space), 52.75 % (3-band tangent space). A small improvement, still far from the paper. Saved in stroke_paper\results\improved_results.txt.

STEP A3 - Export every patient to CSV

```
Script : stroke_paper\export_csv.py
Output : D:\trainsvm\eeg_stroke\csv\  (4.9 GB, readme inside)
  participants.csv         clinical table
  channels.csv             channel number -> name
  sub-XX_raw.csv           raw .mat data, one row per sample (160,000 rows)
  sub-XX_preprocessed.csv  .edf data, same row layout
  sub-XX_events.csv        one row per trial: label, marker positions, quality flags
Columns start with trial, label (1 = left, 2 = right), hand, sample, time_s
(seconds from imagery onset; imagery = 0 to 4 s), then the 33 channels.
Bug found and fixed: some raw marker values are not whole numbers (e.g. 2.8147, after
the break, 10 patients). Rounding them would have created fake "imagery starts". They
are kept as-is and flagged in a column called odd_marker_samples.
```

STEP A4 - Where the paper and the data files disagree

(checked item by item; none of these alone explains the accuracy gap)

- Paralysed side: files say 28 left / 22 right; paper says 27 / 23.
- 2 patients are left-handed (subjects 06 and 15); not highlighted in the paper.
- Average mRS disability score: files 2.68, paper 2.66.
- Paper's usage notes say 20 trials; the files have 40.
- Paper's channel table calls channel 21 "FT7"; it is really TP7.
- Paper's Table 3 lists a "25-25 Hz" band (a typo).
- Channel 32 is "VEOR" in the paper but "HEOR" in the EDF file header.
- Paper says the EDF is split into trials; it is one continuous recording.
- Filtering: paper 8-30 Hz two-way; code 4-30 Hz one-way.
- Testing: paper says 10-fold CV; code does 10 random 60/40 splits.
- A bug in the authors' CSP code: training features use log(variance) but test features use log(1 + variance), so train and test features are on different scales.
- The authors' plotting script calls a function that is missing from the download.
- We downloaded the OLDER version (v1) of the code from figshare to compare. It had extra files (CalIndex.m, PlotAccBar.m, FeatureData.mat), so the published code is not the exact code behind the paper. The archive was password-encrypted and could not be opened, so it was deleted (1.8 GB).

STEP A5 - WHY the paper's numbers cannot be reproduced (the key finding)

Script: stroke_paper\mismatch_analysis.py

Method: re-run each of the authors' test procedures exactly as their code does it, twice:

(a) with the TRUE left/right labels

(b) with the labels RANDOMLY SHUFFLED

With shuffled labels there is nothing real to learn, so any honest method MUST score about 50 %. If a method scores well above 50 % on shuffled labels, its score comes from the procedure, not from the brain.

Procedure (authors' code)	True labels	Shuffled labels	Paper says
FBCSP + SVM	48.27 %	47.05 %	57.57 %
TSLDA alone	53.80 %	47.66 %	-
FgMDM alone	53.90 %	47.77 %	-
TSLDA + FgMDM "fusion"	54.02 %	47.95 %	61.20 %
TWFB + FgMDM	65.95 %	63.79 % <--	72.21 %
FgMDM, one fixed band 8-30 Hz	51.76 %	46.79 %	-

What this shows:

- TWFB + FgMDM (their best, 72 %) still scores 63.8 % when the labels are RANDOM. The reason, from TWFB_DGFMDM.m: it tries 8 frequency bands, each on a new random split, and keeps the band with the HIGHEST TEST accuracy. That is choosing on the test set. The more options you try, the higher the best one scores by luck alone.
- TSLDA + DGFMDRM "fusion" does the same on a smaller scale: in every split it keeps whichever of the two classifiers scored higher ON THE TEST SET.
- Shuffled-label scores come out BELOW 50 % (about 47 %) because the random 24/16 splits are not balanced by class. That is a known side effect of small unbalanced splits.
- Parts of the paper's method (Table 3: time windows, backtracking search, LTSA dimension reduction) are not in the published code at all.
- The EEG signal itself is weak: only 19 of 50 patients show the textbook left/right ERD pattern, which is not more than expected by chance (p = 0.133).

STEP A6 - All models, one honest procedure (the final stroke result)

Script: stroke_paper\run_all_models.py (about 22 minutes, 16 CPU cores)

Input: the CSV files from step A3 (the raw signals)

Honest rules applied to every model:

- stratified 10-fold CV per patient; EXACTLY the same folds for every model, so they can be compared fairly; every trial is tested exactly once
- nothing is ever chosen using test data:
  * "fusion" = average the two classifiers' probabilities (not "pick the winner")
  * TWFB picks its band AND time window by an inner 5-fold CV on training data only
  * SVM C and the selected features are also chosen inside the training data
- two-way (zero-phase) filtering, per-trial offset removal, "shrinkage" covariance estimates (more stable with few trials)

Results (average over 50 patients; "patients above chance" = at least 65 %, i.e. 26 of 40 trials correct, which luck alone reaches with probability under 5 %; with 50 patients, about 2 would pass by pure luck):

Model	Accuracy	Kappa	Patients >= 65 %	Paper
CSP + LDA	50.45 %	0.01	6	55.57 %
FBCSP + SVM (authors' settings)	52.70 %	0.05	10	57.57 %
FBCSP + SVM (tuned)	51.60 %	0.03	8	-
Tangent space + SVM	53.30 %	0.07	9	-
Filter-bank tangent space + SVM	52.75 %	0.05	12	-
TSLDA	53.50 %	0.07	9	-
FgMDM	53.50 %	0.07	9	-
TSLDA + FgMDM fusion	53.50 %	0.07	9	61.20 %
TWFB + FgMDM (band/window chosen on training data only)	55.40 %	0.11	12	72.21 %

How to read it:

- The best honest model is TWFB + FgMDM at 55.4 %. The authors' IDEA (pick a good band and time window per patient) does help a little, when done honestly.
- The best SVM is filter-bank tangent space + SVM (12 patients above chance).
- TSLDA, FgMDM and their fusion give identical answers. That is expected, not a bug: in the library used, FgMDM for two classes ends up using the same shrinkage-LDA on the same tangent-space features.
- Left and right errors are balanced (e.g. TWFB: 550/1000 left and 558/1000 right correct), so no model simply favours one hand.
- A MINORITY of patients really can be decoded, and it is the same ones every time: patients 20 and 40 beat chance with all 9 models; 7, 11 and 23 with 7 of 9. For most patients no method separates left from right.
- Results: stroke_paper\results\ (summary.csv, per_subject_accuracy.csv, confusion_matrices.csv, results_run.log)

(Small note: an early file, improved_results.txt, used 62.5 % as the chance line. The correct line for 40 trials at p < 0.05 is 65 % (26/40), which is used everywhere else.)

PART B - OUR OWN GELO BCI HEADBAND RECORDINGS

4. THE DATA

4.1 Where it is

```
D:\trainsvm\left   5 CSV files: Left1_5, Left6_10, left11_15, left16_20, left21_25
D:\trainsvm\right  10 CSV files: Right1_5, right6_10, ... right46_50
Each file = one recording session holding 5 iterations of the SAME hand.
Recorded on 8, 9 and 12 September.
```

4.2 What each file contains (checked by reading them)

```
Columns: timestamp, ch1 ... ch8, mark
- timestamp goes up by 4000 per row, in microseconds -> 1 row every 4 ms -> 250 Hz.
  (Important: your MATLAB script uses Fs = 256, which is wrong; see section 6.)
- ch1-ch4 are the 4 real electrodes. As you told us: ch2 = C3, ch3 = C4.
  (Your MATLAB script does not state this mapping itself; we took it from you.)
- ch5-ch8 are empty: they hold the constant 1.18e-38 (the smallest float number). As
  you instructed, they are ignored.
- Values are in volts; we convert to microvolts.
- mark:
    1, 2, 3, 4, 5 = the 5 imagery iterations           -> USED
    0             = between iterations                  -> ignored
    6, 7          = about 30 s each, before the task    -> ignored for training
                    (used only as a control test, see 5.5)
```

4.3 Problems found in the recordings (all flagged in the Excel "iterations" sheet)

- Left1_5.csv has NO mark 3, so it has only 4 iterations (you said this is probably a skipped iteration). Total left iterations = 24, not 25.
- Right1_5.csv mark 4 is interrupted by a 0.5 s gap. We treat it as ONE iteration.
- right46_50.csv mark 4 lasts 148 s (others are 5-38 s). We use only its first 40 s (you agreed).
- Every file starts with a large start-up "surge" in the signal (up to about 100,000 times normal power). It is outside the marked iterations, but it is the reason we filter each iteration on its own instead of filtering the whole file.
- Class sizes are unequal: 24 left vs 50 right iterations. We correct for this (5.3).

5. WHAT WE DID WITH THE GELO DATA, STEP BY STEP

STEP B1 - Extract the iterations into Excel

```
Script : gelo_bci\gelo_extract_to_excel.py  (about 45 s)
Output : gelo_bci\output\left.xlsx   - 24 iterations, 125,825 rows
         gelo_bci\output\right.xlsx  - 50 iterations, 273,125 rows
  sheet "data"       one row per sample: label (left/right), iteration (e.g. left_07),
                     file, mark, sample, time_s, ch1-ch4 in microvolts (unfiltered)
  sheet "iterations" one row per iteration: file, mark, number, length, notes
         gelo_bci\output\baseline.csv  the mark 6/7 blocks (for the control test only)
```

```
Iteration numbering follows the file name: in left11_15.csv, mark 1 = left_11,
mark 2 = left_12, and so on.
```

STEP B2 - Prepare the signal for the SVM

(inside gelo_bci\gelo_svm.py)

1. Each iteration separately: remove the offset, remove 50 Hz mains noise, keep 1-40 Hz (two-way Butterworth filter, so no time shift). A second copy is filtered 8-30 Hz (mu + beta) for one of the models.
2. Use at most the first 40 s of each iteration.
3. Cut into 2-second windows, moving 1 s at a time (so neighbouring windows overlap by half). Each window gets its iteration's label. Total: 1,383 windows (466 left, 917 right) from 74 iterations.

STEP B3 - Three SVMs (different features)

```
- svm_c3c4      : ONLY ch2 (C3) and ch3 (C4). Log power in 8-10, 10-13, 13-20,
                  20-30 Hz for each, plus the C3-minus-C4 difference -> 12 numbers.
                  The textbook motor-imagery approach.
- svm_bandpower : all 4 channels. Log power AND relative power in theta 4-8, mu 8-13,
                  low beta 13-20, high beta 20-30 Hz -> 32 numbers.
- svm_tangent   : all 4 channels, 8-30 Hz. The 4x4 covariance matrix (how channels
                  vary together), turned into 10 numbers with the Riemannian tangent
                  space method.
Every SVM:
  - features are scaled first
  - kernel, C and gamma are chosen by an INNER 3-fold CV on training data only
      linear: C = 0.01, 0.1, 1, 10
      rbf   : C = 0.1, 1, 10, 100 and gamma = auto-scaled, 0.01, 0.1
  - class_weight = "balanced" so the larger right class doesn't dominate
Score = BALANCED accuracy (average of the left-correct % and the right-correct %), so
that answering "right" every time scores 50 %, not 68 %.
We also score per ITERATION: each iteration gets the majority vote of its windows.
```

STEP B4 - Three ways of testing (this is where the story is)

```
A. by_iteration : 5 folds; whole iterations are held out (windows of one iteration
                  never split between train and test). BUT the other 4 iterations from
                  the SAME recording file are in training.
B. by_file      : hold out one WHOLE recording file at a time (15 rounds). The model has
                  never seen anything from the recording it is tested on. This is the
                  real-life situation (a new session with the headband), and the only
                  honest test for this data.
C. baseline     : like A, but on the mark 6/7 blocks from BEFORE the task, labelled by
                  the folder they came from. Nobody imagines anything there, so the
                  right answer is 50 %. If it scored high, the model would be recognising
                  the recording, not the imagery.
```

STEP B5 - Results (balanced accuracy; chance = 50 %)

Test	svm_c3c4	svm_bandpower	svm_tangent
A. by_iteration	61.8 % (57.1)	75.4 % (82.6)	77.5 % (81.8)
B. by_file (HONEST)	40.6 % (38.6)	57.3 % (53.0)	65.2 % (66.2)
C. baseline control	52.5 % (57.5)	51.8 % (50.0)	55.4 % (55.0)

(number in brackets = per-iteration majority-vote accuracy)

What it means:

- A looks good (75-78 %), but it falls to 57-65 % in B once the whole recording is unseen. So most of A's accuracy comes from recognising the RECORDING SESSION (electrode contact, noise level, time of day), not the hand. This is possible because every file is only one class: "knowing the session" = "knowing the label".
- C is near 50 %: resting data alone does not give the folder away. So the session effect only shows during the task.
- In B the model tends to label a whole file left or right, all together. So there are really only about 15 independent decisions, not 74. With svm_tangent, about 9 of 15 files come out correct.
- The C3/C4-only model is at or below chance in B (40.6 %). The textbook left/right signal at C3/C4 is not reliable in this data.

STEP B6 - Is 65.2 % real? A permutation test

Script: gelo_bci\gelo_perm.py (about 1.5 hours)

Method: 100 times, randomly re-assign which files are "left" and which are "right" (still 5 left and 10 right), then run the WHOLE procedure B again. This shows what scores luck alone produces with this exact setup.

Result: shuffled average 48.6 %, top 5 % of shuffles 67.2 % or more. 8 of 100 shuffled runs scored 65.2 % or more. p = 0.089 -> NOT significant (it would need to be below 0.05).

Saved: gelo_bci\output\permutation_result.txt and permutation_svm_tangent.csv

(During this test some inner folds had only one class and sklearn printed warnings. Those folds are just skipped; the test still ran correctly to the end.)

STEP B7 - Your MATLAB Ch2/Ch3 ratio script, checked

Your script (eeg_ratio_batch.m): for each iteration, compute the spectrum of Ch2 and Ch3, find the single strongest peak between 0 and 60 Hz, average 6 points around it, report the ratio Ch2/Ch3. Issues found:

1. Fs = 256 is wrong; the data is 250 Hz. Every frequency it reports is 2.4 % too high. Its built-in check only warns above a 5 % difference, so it stays silent. -> set Fs = 250.
2. The "strongest peak" wanders: the peak found ranged from 6.5 Hz (theta) to 17 Hz (beta), with a median of 13 Hz. So it compares different rhythms in different iterations. Fixed bands (mu 8-13 Hz, beta 13-30 Hz) are more meaningful.
3. It splits blocks by contiguous runs, so Right1_5 mark 4 (with its 0.5 s gap) counts as two iterations.

Python replica + fixed-band version (saved: gelo_bci\output\ch2_ch3_ratio_python.csv):

Measure (Ch2 / Ch3)	Left median	Right median	p-value	
Your peak ratio	0.931	0.929	0.95	(no difference)
log mu (8-13 Hz) ratio	-0.14	-0.24	0.049	
log beta (13-30 Hz) ratio	-0.10	-0.05	0.026	

- Mu goes the EXPECTED way (left imagery -> C4 = ch3 weakens -> ratio goes up). This weakly supports ch2 = C3 and ch3 = C4.
- Beta goes the OPPOSITE way from the theory.
- These p-values treat all 74 iterations as independent. They are not (iterations from the same file are alike), so the real evidence is weaker than these numbers suggest. The per-file values of left and right overlap a lot.

STEP B8 - "Combine the data randomly and give it to the model"

Script: gelo_bci\gelo_combined.py

Output: gelo_bci\output\combined_shuffled.xlsx sheet "data" = all 74 iterations, left and right, in random order, with labels sheet "order" = which iteration is at which position gelo_bci\output\combined_results.csv

Test on the mixed data	c3c4	bandpower	tangent
random 10-fold over 2 s windows	68.8 %	85.0 %	85.5 %
(per iteration)	(69.2)	(98.0)	(94.0)
random 10-fold over whole iterations	60.9 %	77.8 %	78.4 %

Why these high numbers can NOT be trusted:

- Random over windows: windows overlap by 1 s, so near-copies of the same signal sit in both train and test (leakage). The model partly just recognises data it has seen.
- Random over iterations: this is the same as test A. Changing the ORDER of the rows does nothing for an SVM. A test iteration still has its 4 sibling iterations from the same file in training.
- Mixing AFTER recording cannot fix the problem. The problem is that each RECORDING contains only one hand. Mixing must happen DURING recording.

PART C - CONCLUSIONS AND WHAT TO DO NEXT

6. CONCLUSIONS

1. Honest testing changes everything. On both datasets, the impressive numbers came from peeking at test data (the paper) or from leakage and a session confound (the headband). The honest numbers are much lower.

2. Stroke dataset: best honest accuracy is 55.4 % (paper: 72.21 %). The paper's extra accuracy is explained by choosing bands/classifiers on the test set. About 12 of 50 patients can be decoded above chance; for the rest, no method works.
3. Gelo headband: on a recording the model has never seen, the best SVM gets 65.2 %, and that is not statistically better than chance (p = 0.089). Left vs right cannot yet be reliably decoded from this data.
4. It is NOT proven that the headband can't do it. The current recordings simply can't tell us, because hand and session are mixed up. Once the design is fixed, the same pipeline will give a clear answer.

7. RECOMMENDATIONS FOR THE NEXT RECORDINGS

1. MOST IMPORTANT: in every session, cue LEFT and RIGHT trials in RANDOM ORDER, with equal numbers of each (e.g. 20 left + 20 right per session). Mark them with different values (e.g. 1 = left, 2 = right), or keep a separate list of the order.
2. Record several sessions on different days, each with both hands. Then test with "leave one session out", which is the honest test.
3. Use short, fixed-length trials, e.g. 2 s cue -> 4 s imagery -> 2-3 s rest, repeated many times. Many short trials beat a few long ones (ours were 5 s to 148 s).
4. Keep the same number of trials for left and right.
5. Keep a rest period (like marks 6/7) in every session; it is a useful control.
6. Check that ch2 and ch3 really sit over C3 and C4 (about 4-5 cm either side of the top centre of the head, over the hand area). Record the exact positions.
7. In MATLAB: set Fs = 250, and use fixed mu (8-13 Hz) and beta (13-30 Hz) bands instead of the "strongest peak".
8. Record with the same person, in a quiet room, without moving, with good electrode contact. Note down anything unusual (like the missing iteration 3 or the 148 s one).

The new data can then be processed by the same scripts (section 8). Only the extraction step needs a small change: take the label from the mark value (or the trial list) instead of from the folder name.

8. FILES AND HOW TO RE-RUN EVERYTHING

```
D:\trainsvm\
  README.md                     short map of the folder
  EXPERIMENT_REPORT.txt         this file
  eeg_stroke\                   stroke dataset (original files, untouched)
    csv\                        our CSV export of every patient (+ README.md)
  left\  right\                 Gelo headband recordings (original files, untouched)
  eeg_headphones\               device SDKs (untouched)

stroke_paper\
  train_fbcsp_svm.py          A1: copy of the authors' FBCSP + SVM          ~1 min
  train_improved_svm.py       A2: improved SVMs                             few min
  export_csv.py               A3: .mat/.edf -> CSV                          ~1.5 min
  mismatch_analysis.py        A5: true vs shuffled labels test              few min
  run_all_models.py           A6: all 9 models, honest CV                   ~22 min
  results\                    summary.csv, per_subject_accuracy.csv,
                              confusion_matrices.csv, results_run.log,
                              improved_results.txt

gelo_bci\
  gelo_extract_to_excel.py    B1: iterations -> left.xlsx / right.xlsx      ~45 s
  gelo_svm.py                 B2-B5: 3 SVMs x 3 tests                        ~6 min
  gelo_combined.py            B8: mix left+right randomly, test              ~5 min
  gelo_perm.py                B6: permutation test (needs gelo_svm.py run)   ~1.5 h
  output\                     all Excel/CSV results listed above
```

Run from a terminal, e.g.: cd D:\trainsvm\gelo_bci python gelo_extract_to_excel.py python gelo_svm.py Software: Python 3.13, numpy, scipy, pandas, scikit-learn 1.7, pyriemann 0.12, joblib, pyarrow, xlsxwriter, openpyxl. All random steps use a fixed seed (0), so results come out the same every time.

9. GLOSSARY (plain-English meanings)

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EEG            Electrical brain activity measured with electrodes on the scalp.
C3 / C4        Standard electrode spots over the left / right hand areas of the brain.
Motor imagery  Imagining a movement without doing it.
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mu / beta	Brain rhythms at 8-13 Hz / 13-30 Hz that weaken during (imagined) movement.
ERD	That weakening ("event-related desynchronisation").
Hz	Cycles per second. Sampling rate 250 Hz = 250 measurements per second.
Filter	Removes unwanted frequencies (e.g. 50 Hz mains hum, slow drift).
Causal / zero-phase filter	Causal = one pass forward, slightly delays the signal (authors' code). Zero-phase = forward and backward, no delay (our code).
Epoch / window	A short slice of signal cut out for analysis (here 2 s or 4 s).
Feature	A number computed from the signal that the classifier uses (e.g. power at C3 in 8-13 Hz).
Band power	How strong the signal is within a frequency range.
Welch PSD	Standard method to estimate power at each frequency.
CSP	Common Spatial Patterns: finds channel mixes that best separate two classes by their power.
FBCSP	Filter-Bank CSP: CSP done separately in several frequency bands.
Covariance matrix	A table of how every pair of channels varies together.
Riemannian / tangent space	A mathematically correct way to turn covariance matrices into ordinary features. It often works better than CSP.
MDM / FgMDM	Classifiers that assign a trial to the class whose "average covariance matrix" is closest (FgMDM adds a filtering step first).
LDA	Linear Discriminant Analysis, a simple linear classifier.
TSLDA	Tangent-space features + LDA.
SVM	Support Vector Machine; finds the boundary between classes with the widest margin.
Kernel	Shape of the SVM boundary: linear (straight) or RBF (curved).
C, gamma	SVM settings: C = how tightly it fits the training data; gamma = how curved an RBF boundary may be.
Scaling	Putting all features on the same scale (mean 0, std 1). SVMs need it.
Training / test set	Data the model learns from / data used only to measure accuracy.
Cross-validation (CV)	Split the data into k parts; train on k-1, test on 1; repeat for all parts.
Stratified	Each fold has the same left/right proportion as the whole.
Grouped CV	Keeps pieces that belong together (same iteration or same file) on the same side of the split.
Nested CV	An inner CV inside the training data, used to choose settings, so the test data is never used for any choice.
Leakage	Test data (or near-copies of it) sneaking into training; gives too-high scores.
Confound	Something besides the thing you study that also differs between classes (here: the recording session).
Test-set selection / peeking	Choosing a setting or model because it scored best on the test data. Makes results look better than they really are.
Chance level	Score from pure guessing: 50 % for two balanced classes.
Balanced accuracy	The average of the per-class accuracies; stays at 50 % for a model that always says the same class.
Kappa	Accuracy corrected for chance: 0 = chance, 1 = perfect.
Confusion matrix	A table counting correct and wrong answers per class.
Permutation test	Shuffle the labels many times and redo everything, to see which scores luck alone produces. p = the share of shuffles that beat the real score.
p-value	The chance of a result this good or better if there were really no effect. Below 0.05 is usually called significant.
Binomial threshold	For 40 trials, 26 or more correct (65 %) happens by luck less than 5 % of the time.